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Integrated molecular pathology and therapeutic outcomes in dermatofibrosarcoma protuberans: Ki-67, COL1A1::PDGFB, and efficacy of Mohs surgery with postoperative imatinib in 153 Chinese patients

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ABSTRACT

Background: Dermatofibrosarcoma protuberans (DFSP) is a rare cutaneous sarcoma with a high local recurrence rate. In one of the largest single-center Asian cohorts (n = 153), we performed integrated pathologic, molecular, and long-term therapeutic analysis to define the prevalence of COL1A1::PDGFB fusion and evaluate the efficacy of Mohs micrographic surgery and postoperative imatinib.

Methods: We retrospectively analyzed 153 consecutive DFSP cases (2018-2023). COL1A1::PDGFB fusion was detected by FISH with reflex RNA sequencing in negative cases; Ki-67 expression was assessed by immunohistochemistry. Treatment outcomes were evaluated in 124 patients with a median follow-up of 26.9 months using descriptive statistics and risk ratio estimates.

Results: Fusion transcripts were identified in 145 of 153 tumours (94.8%). Fibrosarcomatous DFSP showed larger size and higher Ki-67 index than conventional DFSP (5.1 vs 4.1 cm, P = 0.023; 23.4% vs 9.1%, P < 0.001). Among 122 patients followed after MMS, five relapsed (4.1%). Twenty-eight fibrosarcomatous cases received postoperative imatinib for ≥6 months and only two recurred (7.1%). Ki-67 ≥15% strongly correlated with fibrosarcomatous change.

Conclusion: COL1A1::PDGFB fusion remains the molecular hallmark of DFSP. A Ki-67 index ≥15% indicates aggressive biology and should prompt meticulous surgical planning and consideration of postoperative imatinib in high-risk tumours.

Keywords: Dermatofibrosarcoma protuberans, COL1A1::PDGFB, Mohs micrographic surgery, Molecular pathology

1 Introduction

Dermatofibrosarcoma protuberans (DFSP) accounts for <1 % of soft tissue sarcomas and presents as a slow-growing, firm, reddish-brown plaque on the trunk or proximal extremities; rapid expansion may follow trauma [1, 2]. Among histologic variants, fibrosarcomatous DFSP (FS-DFSP) carries a 2- to 3-fold higher risk of local recurrence and distant metastasis [3]. The rarity and morphologic overlap with neurofibroma, keloid, or dermatofibroma mandate ancillary immunohistochemistry and molecular testing [4-6]. COL1A1::PDGFB fusion, identified in up to 95 % of

cases, is the diagnostic hallmark [7, 8]; PDGFD fusions have been reported in approximately 4-5% of DFSP cases and represent a recurrent alternative mechanism [9]. Rare NTRK1/3 rearrangements have also been described, which are diagnostically challenging due to overlap with NTRK-rearranged mesenchymal neoplasms [10-12]. While the diagnostic utility of CD34 and *COL1A1::PDGFB* detection is well-established, practical biomarkers for routine risk stratification remain elusive. Furthermore, the integration of molecular diagnostics with traditional histopathology in guiding therapeutic decisions warrants exploration in large, uniformly managed cohorts. Although MMS reduces local recurrence compared with wide local excision, large-scale integrated pathologic, molecular, and therapeutic data in Asian populations remain scarce [13-15]. Imatinib is effective for advanced disease, but optimal duration and patient selection remain undefined [16, 17].

We therefore conducted a retrospective analysis of 153 consecutive DFSP patients (2018-2023) to delineate clinicopathologic features, genetic landscape, treatment outcomes, and risk factors for recurrence. We hypothesized that RNA sequencing would uncover cryptic fusions in FISH-negative cases and that Ki-67 $\geq 15\%$ would identify high-risk lesions benefiting from imatinib.

2 Materials and methods

2.1 Clinical Cohort and Ethics

We retrospectively enrolled 153 consecutive DFSP patients (2018-2023) diagnosed at the Department of Pathology, Third Affiliated Hospital of Sun Yat-sen University, who underwent primary or post-recurrence Mohs micrographic surgery; 46 patients with prior recurrence were included. Complete case analysis was performed (n=124). Clinical variables comprised gender, age at onset, anatomic site, tumor size, trauma history, histologic subtype, immunohistochemistry, cytogenetics, type of resection, postoperative management, and outcome events (local recurrence or distant metastasis). Data were extracted from electronic medical records and pathology reports.

This study was conducted in accordance with the Declaration of Helsinki and approved by the Institutional Review Board (IRB) of the Third Affiliated Hospital of Sun Yat-sen University (No.2024-089). Informed consent was waived due to the retrospective design. Follow-up information was updated until December 31, 2023; the median follow-up duration was 26.9 months. Postoperative imatinib was offered to patients with high-risk features, defined as fibrosarcomatous DFSP (FS-DFSP) histology, locally recurrent disease, or positive surgical margins after maximal resection. Imatinib was administered orally at 400 mg twice daily. The planned duration was 12 months, with a minimum of 6 months for those who tolerated therapy; early discontinuation was allowed due to adverse events. Local recurrence was defined as histologically confirmed tumor in the operative field after ≥ 1 month of disease-free status. Histological diagnosis of each case was reassessed independently by two dermatopathologists. Cases with insufficient data were ruled out.

2.2 Immunohistochemical Analysis

Immunohistochemistry was performed on 4- μ m formalin-fixed paraffin sections using mouse monoclonal anti-CD34 (clone QBEnd-10, 1:200; MXB Biotechnologies, Ltd. China) and anti-Ki67 (clone MIB-1, 1:100; Zhuhai Quanhui Co., Ltd. China) with automated staining (Ventana BenchMark ULTRA PLUS, Roche Diagnostics, Switzerland). Positive and negative controls were run in parallel. CD34 positivity was defined as >5% cytoplasmic staining. Ki-67 labeling index was quantified in the highest proliferative area (hot-spot) counting 500 tumor cells and expressed as percentage. We randomly selected 20 representative cases and performed automated digital image analysis using QuPath (v0.4.3). The correlation between manual counts and DIA was excellent (Pearson's $r = 0.95$, $P < 0.001$).

2.3 Fluorescence In Situ Hybridization (FISH)

FISH analyses were conducted on paraffin-embedded tissue sections (3.5 μ m thickness) following manufacturer's instructions. A dual-color fusion probe targeting *COL1A1::PDGFB* (ALP, Guangzhou, China) was employed. The specimen was considered positive when >15% of tumor cells displayed a yellow or contiguous red-green signal. The copy number of *COL1A1* and *PDGFB* genes was quantified in 50 cells per tumor for positive cases.

2.4 RNA Sequencing for Fusion Detection

For FISH-negative cases, we performed targeted RNA sequencing to identify cryptic or alternative fusions. Total RNA was extracted from the FFPE tissue samples using AllPrep DNA/RNA FFPE kit as per the manufacturer's procedures (Qiagen, Hilden, Germany). A minimum of 25- μ g total RNA was required for subsequent library preparation. Complementary DNA (cDNA) library preparation involved strand-specific cDNA synthesis, molecular tagging with unique molecular indices (UMI), capture probe hybridization, and gene-specific amplification. Target capture was performed using the OncoRNA panel (Burning Rock Biotech, Guangzhou, China), which covers full transcripts of 115 cancer-related genes, including *PDGFB*, *PDGFRA*, *PDGFRB*, *NTRK1*, *NTRK2*, *NTRK3*, and is designed to identify gene rearrangements and alternative splicing events (see Supplementary Table 1 for the complete gene list). Sequencing was performed using the TruSeq Paired-End (PE) 100 bp kit on the NovaSeq 6000 instrument (Illumina, San Diego, CA, USA). The threshold for sequencing reads was set at > 25 million total reads per sample. Alignment of sequencing reads to the human reference genome (hg19) was performed using STAR (version 2.7.3a) [18]. Fusion transcripts were detected using STAR-Fusion (version 1.8.1) [19].

2.5 Statistical analysis

Continuous variables were tested for normality with Shapiro-Wilk test. Normally distributed data are presented as mean $\pm$ SD and compared using two-tailed independent-samples Student's t-test; non-normally distributed data are expressed as median (interquartile range). Categorical variables were compared with χ^2 test, Fisher's exact test or Fisher-Freeman-Halton exact test for $r \times c$ tables as appropriate. Crude odds ratios (OR) with 95% confidence intervals are reported as exploratory estimates. Due to the low number of recurrence events ($n = 6$), no formal multivariable inference was performed. All analyses were conducted with SPSS 27.0

(IBM, Armonk, NY, USA); two-sided P-values < 0.05 were considered statistically significant.

3. Results

3.1 Clinical characteristics

We retrospectively enrolled 153 consecutive DFSP patients (2018-2023) with a mean age at onset of 29.2 ± 10.1 years (range 0-59). The cohort included 87 males and 66 females (M:F = 1.3:1), with 101/153 (66.0%) aged 20-49 years, including 1 congenital case. Mean tumor size was 4.5 cm (range 0.5-16.0 cm) and 117/153 (76.5%) were trunk lesions. Pre-existing trauma was noted in 17/153 (11.1%).

During a median follow-up of 26.9 months, 6/153 (3.9 %) developed local recurrence and 3/153 (2.0 %) distant metastasis (2 FS-DFSP, 1 C-DFSP), including 1 disease-specific death. 151/153 (98.7 %) underwent Mohs micrographic surgery with clear margins; 2/153 (1.3 %) had wide local excision for metastasis. Imatinib (400 mg bid, 6–12 months) was administered to 30/153 (19.6 %).

3.2 Clinicopathologic Features

Our series comprised 153 DFSP (70 % conventional, 28 % FS-DFSP, one each myoid, mucinous and giant-cell fibroblastoma). Conventional DFSP showed storiform spindle cells with mild atypia and subcutaneous fat infiltration; FS-DFSP exhibited $\geq 5\%$ fibrosarcomatous transformation with mitoses 10-30/10 HPF; myoid variant showed circumscribed eosinophilic nodules; mucinous subtype had $>50\%$ stromal mucin; and giant-cell fibroblastoma contained multinucleated giant cells lining pseudovascular spaces (Figure 1).

3.3 Molecular Features

All 153 tumors showed diffuse CD34 positivity; Ki-67 was positive ($>5\%$) in 131/153 (85.6%) (Figure 2). By FISH, 136 of 153 tumors (88.9%) harbored the *COL1A1::PDGFB* fusion (Figure 3). Eleven FISH-negative cases underwent targeted RNA-seq; nine of them revealed cryptic *COL1A1::PDGFB* transcripts, bringing the final fusion-positive rate to 145/153 (94.8%). Among the remaining three fusion-negative cases, RNA-seq identified rare alternative rearrangements: *AKT3::PDZK1* (E2-E3), *SOAT1::AKT3* (E2-E2) and *TPM3::NTRK1* (E7-E10), each in a single tumor. Based on the hg19 reference genome, the precise breakpoints were as follows: *TPM3::NTRK1* fusion (chr1:154,142,876::chr1:156,844,363) involving *TPM3* (NM_001043353.2) exon 7 and *NTRK1* (NM_001007792.1) exon 10; *AKT3::PDZK1* fusion (chr1:244,006,427::chr1:145,747,042) involving *AKT3* (NM_005465.7) exon 2 and *PDZK1* (NM_002614.4) exon 3; and *SOAT1::AKT3* fusion (chr1:179,271,937::chr1:244,006,584) involving *SOAT1* (NM_001252511.1) exon 2 and *AKT3* (NM_005465.7) exon 2. Co-amplification of *COL1A1* and *PDGFB* was observed in 12 cases. Integrative Genomics Viewer (IGV) plots confirmed the chimeric RNA-seq reads spanning the respective gene breakpoints (Figure 4).

3.4 Clinicopathological Correlations

Clinicopathological comparisons are summarized in Table 2. Male sex was more frequent in FS-DFSP than in C-DFSP (68.8% vs 50.0%, $P=0.031$). Local recurrence occurred in 23/102 (22.5%) C-DFSP vs 21/48 (43.8%) FS-DFSP ($P=0.008$). FS-DFSP

tumors were larger (mean 5.1 vs 4.1 cm, $P=0.023$) and showed higher Ki-67 index (23.4% vs 9.1%, $P<0.001$) than C-DFSP. Using a cutoff of 15%, Ki-67 $\geq 15\%$ was observed in 27/102 (26.5%) of C-DFSP cases and 28/48 (58.3%) of FS-DFSP cases ($P<0.001$). Recurrent tumors were larger (5.3 vs 4.1 cm, $P<0.001$) and had higher Ki-67 (17.9% vs 11.9%, $P=0.007$) than primary tumors (Table 3). Tumor site, *COL1A1::PDGFB* positive rate and gene copy numbers did not differ among subgroups (all $P>0.05$).

3.5 Diagnostic and Prognostic Correlates

The combined use of CD34 immunohistochemistry and *COL1A1::PDGFB* detection yielded a definitive diagnosis in 150/153 cases (98.0%). Among the 11 FISH-negative cases, RNA-seq established the diagnosis in 9 (81.8%), underscoring its value as a reflex test. Ki-67 index $\geq 15\%$ strongly correlated with fibrosarcomatous transformation ($P<0.001$) and showed a trend towards higher recurrence (5.26% vs 4.48%, Table 4). In practical terms, 37/153 cases (24.2%) would be classified as 'high-proliferative' by this cutoff.

3.6 Clinical Management and Outcomes

Of 153 patients, 151/153 (98.7%) achieved clear margins with Mohs micrographic surgery (MMS) and 2/153 (1.3%) underwent wide local excision (WLE). Median follow-up was 26.9 months (range 3.2-55.9). Local recurrence occurred in 5/122 (4.1%) after MMS versus 2/2 (100%) after WLE ($P<0.001$, Table 4); one WLE patient died of disease. Among 33 imatinib-treated patients (29 FS-DFSP, 3 C-DFSP, 1 mucinous type; median 9 months, 400 mg twice daily), recurrence was observed in 4/33 (12.1%). Among these 33 patients, 2 recurrences occurred in those who had received ≥ 6 months of imatinib and 2 recurrences occurred in those who had received <6 months of imatinib. The planned duration of postoperative imatinib was 12 months for all eligible patients. However, the actual treatment duration varied due to imatinib-related adverse events (see Table 5). Among the 33 treated patients, 17 completed the full 12-month course without significant adverse events. Thirteen patients experienced manageable adverse events and discontinued at 6 months as planned. Six patients (18.2%) discontinued treatment prematurely due to severe adverse events (Table 5).

3.7 Risk Factors for Recurrence

In univariate comparisons ($n=153$, 6 events), positive surgical margin (OR 24.2, 95% CI 10.3-57.1, $P=0.049$) and imatinib use (OR 6.2, 95% CI 1.1-35.7, $P=0.043$) were significantly associated with recurrence, whereas age, sex, tumor size, histotype and Ki-67 index showed no significant association (all $P>0.05$). Due to <10 events, these estimates are exploratory and should be viewed as hypothesis-generating.

4 Discussion

This comprehensive analysis of 153 cases of dermatofibrosarcoma protuberans delineates the clinicopathologic spectrum, genetic landscape, and therapeutic outcomes of this entity. Our findings, derived from one of the largest single-institution cohorts to date, not only corroborate key observations from the literature but also unveil novel genetic variants and provide robust evidence to refine clinical

management strategies, particularly regarding the integrated use of Mohs micrographic surgery and imatinib therapy.

4.1 Molecular Landscape of DFSP: Diagnostic Implications

The detection rate of the canonical *COL1A1::PDGFB* fusion gene in our cohort reached 94.8%, aligning with the upper limit of previously reported frequencies and solidifying its role as the molecular hallmark of DFSP [20, 21]. Of significant practical importance was the identification of cryptic fusions by RNA sequencing in 9 of 11 cases that were negative by fluorescence in situ hybridization. This underscores a critical diagnostic pitfall: a negative FISH result in a morphologically suggestive case should not definitively exclude DFSP. Employing more sensitive molecular techniques, such as RNA sequencing, is warranted in such scenarios to prevent diagnostic delay or misclassification. This aligns with the growing adoption of RNA-seq as a primary diagnostic tool for detecting gene fusions in sarcomas, offering superior sensitivity and the ability to detect novel partners [22, 23].

More innovatively, our study identified several rare non-*COL1A1::PDGFB* fusions, expanding the molecular spectrum of DFSP. The discovery of a *TPM3::NTRK1* fusion is of particular translational significance. NTRK fusions are oncogenic drivers across multiple tumor types, and selective TRK inhibitors (larotrectinib, entrectinib) have demonstrated remarkable efficacy, with objective response rates exceeding 75% in NTRK fusion-positive solid tumors regardless of histology [17, 24, 25]. In the specific context of sarcomas, TRK inhibitors have shown durable responses in NTRK-rearranged soft tissue sarcomas [26, 27]. While DFSP is typically driven by *COL1A1::PDGFB* and responsive to imatinib, resistance can develop. The presence of an NTRK fusion provides a rationale for alternative targeted therapy in imatinib-resistant or refractory DFSP. Clinically, for a DFSP patient who progresses on imatinib or harbors an NTRK fusion as the primary driver, referral for TRK inhibitor therapy may be considered. This finding underscores a critical principle in modern sarcoma management: the increasing necessity to screen for histology-agnostic therapeutic targets, even in entities where they are exceptionally rare. Indeed, recent diagnostic guidelines have begun to incorporate NTRK fusion testing for sarcomas, recognizing its role in guiding targeted therapy [26, 27]. Our case provides a concrete rationale for considering such testing in DFSP, especially in aggressive or refractory cases.

Furthermore, we report for the first time the novel *AKT3::PDZK1* and *SOAT1::AKT3* fusions in DFSP. AKT3 is a central node in the oncogenic PI3K/AKT/mTOR pathway, governing cell growth and survival[28]. To our knowledge, these fusions have not been previously described in DFSP or any other neoplasm.

The clinical and pathological features of the two index cases are summarized as follows. The patient with *AKT3::PDZK1* fusion was a 41-year-old female with fibrosarcomatous DFSP (Ki-67 20%) arising from the right shoulder. The tumor had an indolent course over 20 years, with first recurrence 9 years after initial surgery and second recurrence 3 years later. She did not receive imatinib and has remained free of

disease for over 36 months after the most recent surgery. The patient with SOAT1::AKT3 fusion was a 41-year-old male with conventional DFSP (Ki-67 25%) located on the trunk. He received postoperative imatinib (0.4 g once daily) and has shown no evidence of recurrence.

Although functional studies are required to confirm whether these rearrangements lead to constitutive AKT3 activation, they represent compelling candidates for novel oncogenic mechanisms and potential therapeutic targets, meriting further investigation in future studies.

4.2 Ki-67 as a Practical Adjunct in DFSP Risk Stratification

Our data confirm that fibrosarcomatous transformation is associated with larger tumor size, a higher Ki-67 index, and a trend toward increased recurrence. This aligns with the established role of high-grade transformation as a key prognostic determinant [29]. Importantly, we extend this observation by demonstrating that Ki-67 itself is a continuous biomarker of intrinsic aggressiveness: recurrent tumors harbored significantly higher indices than primary tumors.

Within this context, a cutoff of $\geq 15\%$ identified a distinct “high-proliferative” group (57/153, 37.3% of our cohort), which correlated strongly with adverse features like larger size and FS-DFSP histology. While the statistical power to link this binary cutoff directly to recurrence was limited in our cohort, its consistent association with validated markers of aggressiveness supports its correlative utility. This effort to define actionable Ki-67 thresholds is part of a broader translational endeavor in sarcoma pathology, as reflected in recent studies [30, 31].

Therefore, we propose that a Ki-67 index $\geq 15\%$ should be routinely reported and interpreted as a practical “red flag” for heightened clinical vigilance. It serves to flag tumors with aggressive biological potential, prompting consideration of wider surgical margins or intensified surveillance within a multidisciplinary framework, even in the absence of overt FS-DFSP morphology. Prospective validation of this threshold is warranted.

4.3 Therapeutic Outcomes and Clinical Management

The therapeutic follow-up data from our study provide compelling real-world evidence for DFSP management. Among 122 patients with follow-up data who underwent MMS, the overall recurrence rate was remarkably low at 4.1%, with a rate of only 1.2% for conventional DFSP. This outcome is superior to the recurrence rates typically reported for wide local excision (often 10–20%) and is consistent with recent meta-analyses [32, 33]. Our results suggest that MMS may be preferred as the first-line surgical modality for primary or locally recurrent DFSP where anatomically feasible.

For FS-DFSP, which carries a substantially higher risk of recurrence and metastasis, surgery alone may be insufficient [34, 35]. Our study offers valuable insights in this challenging area. In the subset of 28 FS-DFSP patients who received postoperative imatinib (for 6–12 months) following MMS, the recurrence rate was 7.1%. While derived from a retrospective analysis, this encouraging figure is congruent with

smaller series exploring imatinib in the adjuvant setting [36]. As a platelet-derived growth factor receptor inhibitor, imatinib directly targets the oncogenic pathway driven by the *COL1A1::PDGFB* fusion [37, 38]. Our experience suggests that postoperative imatinib is a rational and effective consolidative strategy for high-risk FS-DFSP patients who have undergone grossly complete resection.

4.4 Limitations and Future Directions

Our study has several inherent limitations. First, its single-center, retrospective design may introduce selection bias. The proportion of FS-DFSP in our cohort (31.4%) is higher than the commonly cited 10-15%, potentially reflecting the referral of more complex or advanced cases to our tertiary center. Second, while sufficient for assessing early recurrence, the median follow-up of approximately 27 months remains relatively short for a tumor known for late recurrences. Longer follow-up is needed to confirm the durability of the therapeutic outcomes. Additionally, our targeted RNA-seq panel (OncoRNA) does not include probes for PDGFD. Given that PDGFD fusions account for approximately 4-5% of DFSP cases, our fusion detection rate (94.8%) may slightly underestimate the true prevalence of actionable fusions in this cohort. Future studies employing comprehensive whole-transcriptome or DNA-based sequencing are needed to fully capture the spectrum of DFSP-associated rearrangements.

Before these observations can change everyday practice, several questions must be addressed. Prospective multicenter studies are needed to validate Ki-67 values and molecular signatures into a clinically applicable risk score. For patients with NTRK1 fusions, the efficacy of TRK inhibition requires further evaluation. Finally, randomized trials should define the optimal duration of imatinib and whether its benefit extends to patients with classic DFSP but narrow margins or repeated local relapse.

5 CONCLUSION

COL1A1::PDGFB fusion remains the genetic hallmark of DFSP, and Ki-67 $\geq 15\%$ identifies lesions with accelerated growth that merit consideration of wider margins or postoperative imatinib. Mohs micrographic surgery delivers excellent local control, whereas postoperative imatinib may be considered for fibrosarcomatous or otherwise high-risk tumors in a clinical trial or multidisciplinary setting, pending further validation. These findings underscore the need for prospective clinical trials to validate the Ki-67 threshold, define the optimal duration of postoperative imatinib, and directly compare imatinib against observation or radiotherapy in high-risk, completely resected DFSP.

List of abbreviations

C-DFSP: Conventional dermatofibrosarcoma protuberans

cDNA: Complementary DNA

CI: Confidence interval

DFS: Disease-free survival

DFSP: Dermatofibrosarcoma protuberans
FFPE: Formalin-fixed paraffin-embedded
FISH: Fluorescence in situ hybridization
FS-DFSP: Fibrosarcomatous dermatofibrosarcoma protuberans
HPF: High-power field
H&E: Hematoxylin and eosin
ICC: Intraclass correlation coefficient
IGV: Integrative Genomics Viewer
IRB: Institutional Review Board
MMS: Mohs micrographic surgery
NGS: Next-generation sequencing
OR: Odds ratio
OS: Overall survival
PDGF: Platelet-derived growth factor
PE: Paired-end
RFS: Recurrence-free survival
RNA-seq: RNA sequencing
RT-PCR: Reverse transcription polymerase chain reaction
SD: Standard deviation
UMI: Unique molecular index
WLE: Wide local excision

Declarations

1.Ethics approval and consent to participate

The study was conducted in accordance with the Declaration of Helsinki and approved by the Institute Research Ethics Committee of the Third Affiliated Hospital of Sun Yat-sen University (No.2024-089). Written informed consent was obtained from all individual participants.

2.Consent for publication

Not applicable.

3.Availability of data and materials

The datasets generated and/or analysed during the current study are available in the Genome Sequence Archive (Genomics, Proteomics & Bioinformatics 2025) repository, <https://ngdc.cncb.ac.cn/gsa-human> (accession HRA016372).

4.Competing interests

The authors declare that they have no competing interests.

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manuscript.

6. Authors' contributions

Jinrui Guo was responsible for study conception and design, data analysis, and drafting of the manuscript.

Yongmei Cui: Investigation, Methodology.

Nana Zhang: Resources, Validation.

Jing Liang: Visualization, Data curation.

Jian Ning Chen: Histopathological diagnosis and immunohistochemistry interpretation, Validation.

Xiangqi Huang: Writing–review & editing (multiple rounds of critical revision, logical restructuring and final approval of the manuscript) and co-correspondence with the journal.

Dan He: oversaw the study and granted final approval for the manuscript's submission.

7. Acknowledgements

Not applicable.

References:

1. Saiag P, Grob J, Lebbe C, Malvehy J, Del Marmol V, Pehamberger H, Peris K, Stratigos A, Middleton M, Basholt L *et al*: **Diagnosis and treatment of dermatofibrosarcoma protuberans. European consensus-based interdisciplinary guideline.** *EUR J CANCER* 2015, **51**(17):2604-2608.
2. Acosta AE, Velez CS: **Dermatofibrosarcoma Protuberans.** *CURR TREAT OPTION ON* 2017, **18**(9):56.
3. Thway K, Noujaim J, Jones RL, Fisher C: **Dermatofibrosarcoma protuberans: pathology, genetics, and potential therapeutic strategies.** *ANN DIAGN PATHOL* 2016, **25**:64-71.
4. Liang CA, Jambusaria-Pahlajani A, Karia PS, Elenitsas R, Zhang PD, Schmults CD: **A systematic review of outcome data for dermatofibrosarcoma protuberans with and without fibrosarcomatous change.** *J AM ACAD DERMATOL* 2014, **71**(4):781-786.
5. Reha J, Katz SC: **Dermatofibrosarcoma Protuberans.** *SURG CLIN N AM* 2016, **96**(5):1031-1046.
6. Zhang Z, Chen H, Chen M, He X, Wang Y, Zhang H: **Application of COL1A1-PDGFB fusion gene detection by fluorescence in situ hybridization in biopsy tissue of dermatofibrosarcoma protuberans.** *J DERMATOL* 2017, **44**(7):798-802.
7. Allen A, Ahn C, Sanguenza OP: **Dermatofibrosarcoma Protuberans.** *DERMATOL CLIN* 2019, **37**(4):483-488.
8. Karanian M, Perot G, Coindre J, Chibon F, Pedeutour F, Neuville A: **Fluorescence in situ hybridization analysis is a helpful test for the diagnosis of dermatofibrosarcoma protuberans.** *MODERN PATHOL* 2015, **28**(2):230-237.
9. Sotiriou S, Chatzopoulos K, Charville GW, Bean GR, Michal M, Gross JM, Bridge JA, Gardner JM, Agaimy A, Ng TL *et al*: **Dermatofibrosarcoma protuberans with PDGFD rearrangements: a case series featuring a novel EMILIN1::PDGFD fusion and comprehensive literature review.** *VIRCHOWS ARCH* 2025, **487**(2):349-361.

10. Malan M, Xuejingzi W, Quan SJ: **The efficacy of Mohs micrographic surgery over the traditional wide local excision surgery in the cure of dermatofibrosarcoma protuberans.** *PAN AFR MED J* 2019, **33**:297.
11. Martin ECS, Vyas KS, Batbold S, Erwin PJ, Brewer JD: **Dermatofibrosarcoma Protuberans Recurrence After Wide Local Excision Versus Mohs Micrographic Surgery: A Systematic Review and Meta-Analysis.** *DERMATOL SURG* 2022, **48**(5):479-485.
12. Olson N, Rouhi O, Zhang L, Angeles C, Bridge J, Lopez-Terrada D, Royce T, Linos K: **A novel case of an aggressive superficial spindle cell sarcoma in an adult resembling fibrosarcomatous dermatofibrosarcoma protuberans and harboring an EML4-NTRK3 fusion.** *J CUTAN PATHOL* 2018, **45**(12):933-939.
13. Hao X, Billings SD, Wu F, Stultz TW, Procop GW, Mirkin G, Vidimos AT: **Dermatofibrosarcoma Protuberans: Update on the Diagnosis and Treatment.** *J CLIN MED* 2020, **9**(6).
14. McArthur GA, Demetri GD, van Oosterom A, Heinrich MC, Debiec-Rychter M, Corless CL, Nikolova Z, Dimitrijevic S, Fletcher JA: **Molecular and clinical analysis of locally advanced dermatofibrosarcoma protuberans treated with imatinib: Imatinib Target Exploration Consortium Study B2225.** *J CLIN ONCOL* 2005, **23**(4):866-873.
15. van der Graaf WTA, Gelderblom H: **New systemic therapy options for advanced sarcomas.** *CURR TREAT OPTION ON* 2012, **13**(3):306-317.
16. Zheng Z, Piao J, Lee J, Kim S, Kim S, Chung KY, Roh MR: **Dermatofibrosarcoma protuberans: a study of clinical, pathologic, genetic, and therapeutic features in Korean patients.** *YONSEI MED J* 2015, **56**(2):440-446.
17. Llombart B, Monteagudo C, Sanmartin O, Lopez-Guerrero JA, Serra-Guillen C, Poveda A, Jorda E, Fernandez-Serra A, Pellin A, Guillen C *et al*: **Dermatofibrosarcoma protuberans: a clinicopathological, immunohistochemical, genetic (COL1A1-PDGFB), and therapeutic study of low-grade versus high-grade (fibrosarcomatous) tumors.** *J AM ACAD DERMATOL* 2011, **65**(3):564-575.
18. Dobin A, Davis CA, Schlesinger F, Drenkow J, Zaleski C, Jha S, Batut P, Chaisson M, Gingeras TR: **STAR: ultrafast universal RNA-seq aligner.** *BIOINFORMATICS* 2013, **29**(1):15-21.
19. Haas BJ, Dobin A, Li B, Stransky N, Pochet N, Regev A: **Accuracy assessment of fusion transcript detection via read-mapping and de novo fusion transcript assembly-based methods.** *GENOME BIOL* 2019, **20**(1):213.
20. Peng R, Zhang G, Li H: **COL1A1-PDGFB Fusion Gene Detection Through Bulk RNA-Seq and Transcriptomic Features of Dermatofibrosarcoma Protuberans.** *DERMATOL SURG* 2023, **49**(5S):S27-S33.
21. Zhang Z, Lu Y, Shi C, Chen M, He X, Zhang H: **Pediatric dermatofibrosarcoma protuberans: A clinicopathologic and genetic analysis of 66 cases in the largest institution in Southwest China.** *FRONT ONCOL* 2023, **13**:1017154.
22. Horak P, Heining C, Kreutzfeldt S, Hutter B, Mock A, Hullein J, Frohlich M, Uhrig S, Jahn A, Rump A *et al*: **Comprehensive Genomic and Transcriptomic Analysis for Guiding Therapeutic Decisions in Patients with Rare Cancers.** *CANCER DISCOV* 2021, **11**(11):2780-2795.
23. Hetherington E, Adhikari K, Scime NV, Metcalfe A: **Cesarean Deliveries Among Immigrant and Canadian-Born Women in a Representative Community Population in Canada: A Retrospective Cohort Study.** *J OBSTET GYNAECOL CA* 2022, **44**(2):148-156.
24. Hsiao SJ, Zehir A, Sireci AN, Aisner DL: **Detection of Tumor NTRK Gene Fusions to Identify**

Patients Who May Benefit from Tyrosine Kinase (TRK) Inhibitor Therapy. *J MOL DIAGN* 2019, **21**(4):553-571.

25. Frehner L, Haeffliger S, Cerciello F, Grob T, Schmid S: **Complete Response on Larotrectinib in NTRK2 Fusion-Positive Non-Small Cell Lung Cancer.** *CASE REP ONCOL* 2023, **16**(1):871-877.

26. Hamran S, Alnoubani Z, Hassan AM, Al-Rajhi AA, Rajha HE, Altaissan M, Babu GR, Gatalica Z, Vranic S: **NTRK1-3 fusions in sarcomas: prevalence, significance, and clinical implications - a systematic review.** *FUTURE ONCOL* 2025, **21**(30):4011-4024.

27. Hasegawa N, Hayashi T, Niizuma H, Kikuta K, Imanishi J, Endo M, Ikeuchi H, Sasa K, Sano K, Hirabayashi K *et al*: **Detection of Novel Tyrosine Kinase Fusion Genes as Potential Therapeutic Targets in Bone and Soft Tissue Sarcomas Using DNA/RNA-based Clinical Sequencing.** *CLIN ORTHOP RELAT R* 2024, **482**(3):549-563.

28. Prabhu KS, Kuttikrishnan S, Mariyam Z, Habeeba U, Panicker AJ, Masoodi T, Junejo K, Uddin S: **PI3 K/AKT/mTOR pathway and its role in breast cancer stem cells.** *N-S ARCH PHARMACOL* 2025, **398**(12):16779-16795.

29. Llombart B, Monteagudo C, Sanmartín O, López-Guerrero JA, Serra-Guillén C, Poveda A, Jorda E, Fernandez-Serra A, Pellín A, Guillén C *et al*: **Dermatofibrosarcoma protuberans: a clinicopathological, immunohistochemical, genetic (COL1A1-PDGFB), and therapeutic study of low-grade versus high-grade (fibrosarcomatous) tumors.** *J AM ACAD DERMATOL* 2011, **65**(3):564-575.

30. Laurila RE, Bohling TO, Blomqvist CP, Karlsson C, Tukiainen EJ, Repo J, Sampo MM: **Visual Counting and Automated Image-analytic Assessment of Ki-67 and their Prognostic Value in Synovial Sarcoma.** *Cancer Diagn Progn* 2022, **2**(1):7-14.

31. Meng Y, Quan Q, Zhang F, Liu Y, Ren S, Mu X: **Prognostic Value of Ki-67 Index in Patients With Endometrial Stromal Sarcoma.** *FRONT MED-LAUSANNE* 2021, **8**:823505.

32. Martin ECS, Vyas KS, Batbold S, Erwin PJ, Brewer JD: **Dermatofibrosarcoma Protuberans Recurrence After Wide Local Excision Versus Mohs Micrographic Surgery: A Systematic Review and Meta-Analysis.** *Dermatologic surgery : official publication for American Society for Dermatologic Surgery [et al.]* 2022, **48**(5):479-485.

33. Hao X, Billings SD, Wu F, Stultz TW, Procop GW, Mirkin G, Vidimos AT: **Dermatofibrosarcoma Protuberans: Update on the Diagnosis and Treatment.** *J CLIN MED* 2020, **9**(6):1752.

34. Remiszewski P, Pisklak A, Filipek K, Spalek MJ, Szumera-Cieckiewicz A, Szostakowski B, Krotewicz M, Czarnecka AM: **Dermatofibrosarcoma Protuberans (DFSP): Current Treatments and Clinical Trials.** *CURR TREAT OPTION ON* 2025, **26**(11):967-989.

35. Bordeaux J, Blitzblau R, Aasi SZ, Alam M, Amini A, Bibee K, Bolotin D, Chen P, Contreras CM, DiMaio D *et al*: **Dermatofibrosarcoma Protuberans, Version 1.2025, NCCN Clinical Practice Guidelines In Oncology.** *J NATL COMPR CANC NE* 2025, **23**(1).

36. McArthur GA, Demetri GD, van Oosterom A, Heinrich MC, Debiec-Rychter M, Corless CL, Nikolova Z, Dimitrijevic S, Fletcher JA: **Molecular and clinical analysis of locally advanced dermatofibrosarcoma protuberans treated with imatinib: Imatinib Target Exploration Consortium Study B2225.** *J CLIN ONCOL* 2005, **23**(4):866-873.

37. Rota S, Franza A, Fabbri C, Paolini B, Greco FG, Alessi A, Padovano B, Casali P, Sanfilippo R: **COL1A1::PDGFB fusion-associated uterine fibrosarcoma: A case report and review of the literature.** *CANCER REP-US* 2024, **7**(2):e1969.

38. Singh H, Choudhary HB, Mandlik DS, Magre MS, Mohanto S, Ahmed MG, Singh BK, Mishra AK,

Kumar A, Mishra A *et al*: **Molecular pathways and therapeutic strategies in dermatofibrosarcoma protuberans (DFSP): unravelling the tumor's genetic landscape.** *EXCLI J* 2024, **23**:727-762.

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Tables

Table 1 Baseline clinicopathologic characteristics of 153 DFSP patients

Characteristic	Value
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Characteristic	Value
Demographics	
Total patients, n	153
Age at onset (years), mean $\pm$ SD (range)	29.2 $\pm$ 10.1 (0 - 59)
Gender, n (%)	
Male	87 (56.9%)
Female	66 (43.1%)
Male-to-female ratio	1.3:1
Tumor characteristics	
Tumor size (cm), mean $\pm$ SD (range)	4.5 $\pm$ 2.8 (0.5 - 16)
Trauma history, n (%)	17 (11.1%)
Anatomic location, n (%)	
Head and neck	14 (9.2%)
Extremities	22 (14.4%)
Trunk	117 (76.5%)
Histologic subtype, n (%)	
Conventional DFSP	102 (66.7%)
Fibrosarcomatous DFSP (FS-DFSP)	48 (31.4%)
Mucinous DFSP	1 (0.65%)
Myoid DFSP	1 (0.65%)
Giant cell fibroblastoma	1 (0.65%)
Prior treatment history	
Prior recurrence (before study entry), n (%)	46 (30.1%)
Surgical management	
Mohs micrographic surgery (MMS), n (%)	151 (98.7%)
Wide local excision (WLE), n (%)	2 (1.3%)
Postoperative imatinib* (≥ 6 months, 400 mg bid) , n (%)	33 (21.6%)
Follow-up outcomes (median 26.9 months)	
Local recurrence during follow-up, n (%)	6 (3.9%)
Distant metastasis, n (%)	3 (2.0%)

*Postoperative imatinib was administered at 400 mg twice daily. Use in this setting is off-label and exploratory.

Table 2 Clinicopathological comparison between conventional DFSP and fibrosarcomatous DFSP

Characteristic	C-DFSP (n=102)	FS-DFSP (n=48)	t/χ^2	P value
Age, years, mean $\pm$ SD	29.4 $\pm$ 10.6	29.7 $\pm$ 11.8	-0.139	0.135
Sex, Male, n (%)	51 (50.0%)	33 (68.8%)	4.657	0.031
Tumor Size, cm, mean $\pm$ SD	4.1 $\pm$ 2.3	5.1 $\pm$ 3.0	-2.302	0.023
Recurrence, n (%)	23 (22.5 %)	21 (43.8 %)	7.078	0.008
Ki-67 index, %, mean $\pm$ SD	9.1 $\pm$ 5.8	23.35 $\pm$ 12.7	-7.406	<0.001
Ki-67 $\geq$ 15%, n (%)	27 (26.5%)	28 (58.3%)	14.270	<0.001
CD34 positive, n (%)	102 (100%)	48 (100%)		
COL1A1::PDGFB Fusion, n (%)	97 (95.1%)	46 (95.8%)	0.04	0.99
PDGFB copy number, mean $\pm$ SD	2.52 $\pm$ 0.56	2.45 $\pm$ 0.52	0.73	0.47
COL1A1 copy number, mean $\pm$ SD	2.57 $\pm$ 0.46	2.59 $\pm$ 0.51	-0.24	0.81
Tumor Location, n (%)			3.008	0.222
Head & Neck	10 (9.8%)	3 (6.3%)		
Extremities	11 (10.8%)	10 (20.8%)		
Trunk	81 (79.4%)	35 (72.9%)		

Table 3 Clinicopathological comparison between primary and recurrent tumors

Characteristic	Primary (n=107)	Recurrence (n=46)	t/χ^2	P value
Age, years, mean $\pm$ SD	29.64 $\pm$ 10.5	28.2 $\pm$ 12.5	0.709	0.479
Sex, Male, n (%)	59 (55.1%)	28 (60.9%)	0.431	0.512
Tumor Size, cm, mean $\pm$ SD	4.109 $\pm$ 2.256	5.296 $\pm$ 3.091	-2.655	0.009
Ki67 Index, %, mean $\pm$ SD	11.93 $\pm$ 9.065	17.91 $\pm$ 13.366	-2.772	0.007
CD34 mean $\pm$ SD	107 (100%)	46 (100%)	-	-
Tumor Location, n (%)			0.486	0.784
Head & Neck	10 (9.3%)	4 (8.7%)		
Extremities	14 (13.1%)	8 (17.4%)		
Trunk	83 (77.6%)	34 (73.9%)		

Table 4 Univariate analysis of risk factors for local recurrence in 124 DFSP patients

Variable	Category	n	Recurrence n (%)	Crude OR (95% CI)‡	P value†
Age	<30 y	69	4(5.80)	1.63 (0.29-9.25)	0.69
	≥30 y	55	4(5.80)	1 (ref)	
Sex	Male	69	4(5.80)	0.61 (0.11-3.48)	0.69
	Female	55	2(3.64)	1 (ref)	
Surgical modality	WLE	2	2(100)	81.0 (3.5–1876)	0.003
	MMS	122	5(4.1)	1 (ref)	-
Tumor size	<3 cm	34	3(8.82)	2.81 (0.54-14.64)	0.344
	≥3 cm	90	3(3.33)	1 (ref)	
Histotype	FS-DFSP	40	3(7.5)	3.3 (0.5–21.1)	0.21
	C-DFSP	82	2(2.44)	1 (ref)	-
	Others*	2	1(50.0)	-	-
Ki67 index	≥15%	57	3(5.26)	0.844 (0.16-4.35)	0.99
	<15	67	3(4.48)	1 (ref)	-
Margin status	positive	1	1(100)	24.2 (10.26-57.09)	0.049
	negative	121	5(4.13)	1 (ref)	
Imatinib use	Yes	33	4(12.12)	0.163 (0.03-0.94)	0.043
	NO	91	2(2.06)	1 (ref)	-

*Others: mucinous DFSP (n = 1) and giant-cell fibrosarcoma (n = 1).

† Fisher's exact test (2 groups) or Fisher-Freeman-Halton exact test (> 2 groups).

‡ Exploratory estimates due to only six events; blank cells = uncomputable.

§ OR for positive margin is explorative due to n = 1 and 100 % recurrence; exact 95 % CI is highly unstable.

Table 5 Adverse events in imatinib-treated patients (n = 33)

Event	n
Any adverse event	17(51.5%)
Discontinuation due to severe adverse events	6(18.2%)
Gastrointestinal symptoms	6(18.2%)
Periorbital edema	4(12.1%)
Skin rash	4(12.1%)
Hepatic/renal dysfunction	2(6.1%)

Supplementary Table 1. Gene composition of the OncoRNA panel

No.	Gene Symbol	No.	Gene Symbol	No.	Gene Symbol
1	<i>ABL1</i>	41	<i>FGR</i>	81	<i>PATZ1</i>
2	<i>AKT3</i>	42	<i>FLI1</i>	82	<i>PAX3</i>
3	<i>ALK</i>	43	<i>FLT1</i>	83	<i>PAX7</i>
4	<i>AR</i>	44	<i>FLT3</i>	84	<i>PDGFB</i>
5	<i>ARHGAP26</i>	45	<i>FOXO1</i>	85	<i>PDGFRA</i>
6	<i>ASPSCR1</i>	46	<i>FOXO4</i>	86	<i>PDGFRB</i>
7	<i>ATF1</i>	47	<i>FUS</i>	87	<i>PIK3CA</i>
8	<i>AXL</i>	48	<i>GLI1</i>	88	<i>PKN1</i>
9	<i>BCL2</i>	49	<i>HMGA2</i>	89	<i>PLAG1</i>
10	<i>BRAF</i>	50	<i>INSR</i>	90	<i>PPARG</i>
11	<i>BRCA1</i>	51	<i>JAK2</i>	91	<i>PRKCA</i>
12	<i>BRCA2</i>	52	<i>JAZF1</i>	92	<i>PRKCB</i>
13	<i>BRD3</i>	53	<i>KDR</i>	93	<i>RAF1</i>
14	<i>BRD4</i>	54	<i>KIF5B</i>	94	<i>RELA</i>
15	<i>CAMTA1</i>	55	<i>KIT</i>	95	<i>RET</i>
16	<i>CCNB3</i>	56	<i>KMT2A</i>	96	<i>ROS1</i>
17	<i>CDK4</i>	57	<i>MAML2</i>	97	<i>RPS6KB1</i>
18	<i>CIC</i>	58	<i>MAST1</i>	98	<i>RSPO2</i>
19	<i>COL1A1</i>	59	<i>MAST2</i>	99	<i>RSPO3</i>
20	<i>CREB1</i>	60	<i>MEAF6</i>	100	<i>SS18</i>
21	<i>CREB3L2</i>	61	<i>MET</i>	101	<i>SSX1</i>
22	<i>CSF1R</i>	62	<i>MLLT3</i>	102	<i>SSX2</i>
23	<i>DDIT3</i>	63	<i>MRTFB</i>	103	<i>SSX4</i>
24	<i>EGFR</i>	64	<i>MSH2</i>	104	<i>STAT6</i>
25	<i>EML4</i>	65	<i>MSMB</i>	105	<i>TAF15</i>
26	<i>EPC1</i>	66	<i>MUSK</i>	106	<i>TCF12</i>
27	<i>ERBB2</i>	67	<i>MYB</i>	107	<i>TERT</i>
28	<i>ERG</i>	68	<i>MYC</i>	108	<i>TFE3</i>
29	<i>ESR1</i>	69	<i>NCOA1</i>	109	<i>TFEB</i>
30	<i>ETS1</i>	70	<i>NCOA2</i>	110	<i>TFG</i>
31	<i>ETV1</i>	71	<i>NOTCH1</i>	111	<i>THADA</i>
32	<i>ETV4</i>	72	<i>NOTCH2</i>	112	<i>TMPRSS2</i>
33	<i>ETV5</i>	73	<i>NOTCH3</i>	113	<i>USP6</i>
34	<i>ETV6</i>	74	<i>NR4A3</i>	114	<i>WT1</i>
35	<i>EWSR1</i>	75	<i>NRG1</i>	115	<i>YWHAE</i>
36	<i>FEV</i>	76	<i>NTRK1</i>		
37	<i>FGFR1</i>	77	<i>NTRK2</i>		
38	<i>FGFR2</i>	78	<i>NTRK3</i>		
39	<i>FGFR3</i>	79	<i>NUMBL</i>		
40	<i>FGFR4</i>	80	<i>NUTM1</i>		

Figure legends

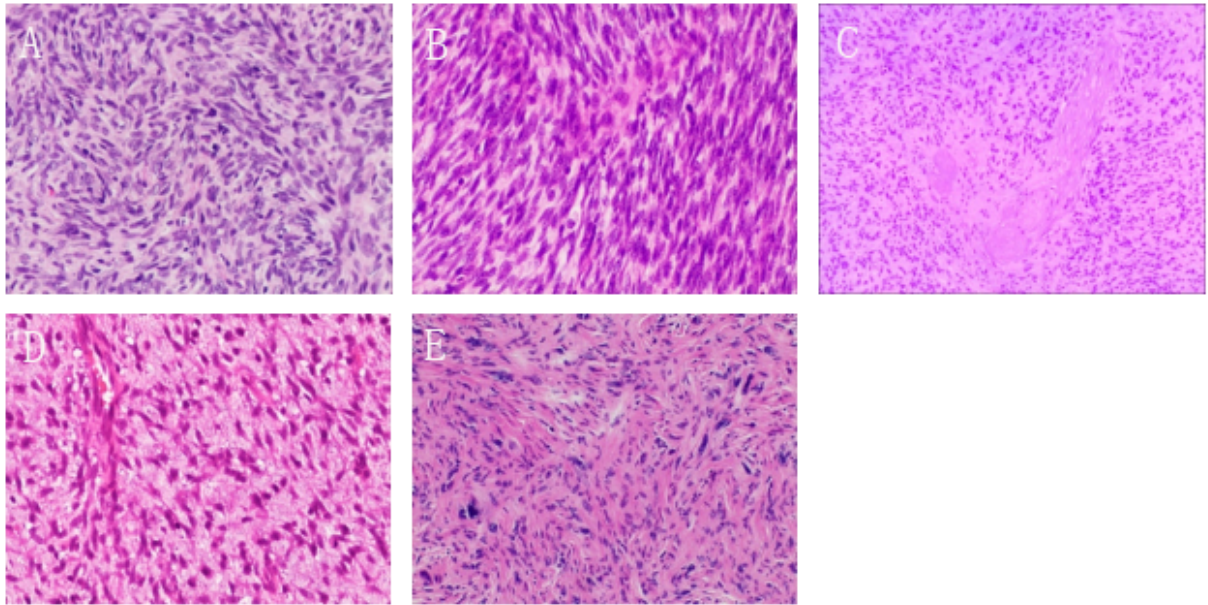


Figure 1 Morphologic variants of dermatofibrosarcoma protuberans (DFSP).

(A) Conventional DFSP: storiform spindle cells with mild atypia and subcutaneous fat infiltration (H&E, 100×).

(B) FS-DFSP: ** ≥ 5 % fibrosarcomatous transformation with mitoses 10–30/10 HPF (H&E, 100×).

(C) Myoid variant: ** circumscribed eosinophilic nodules (H&E, 40×).

(D) Mucinous subtype: ** >50 % stromal mucin with floating spindle cells (H&E, 100×).

(E) Giant-cell fibroblastoma: ** multinucleated giant cells lining pseudovascular spaces (H&E, 100×).

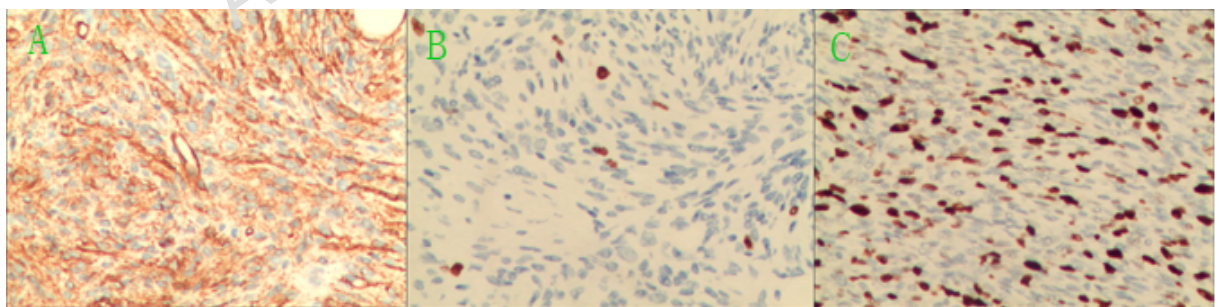


Figure 2 Immunohistochemical staining for CD34 and Ki-67, respectively.

(A) The spindle cells exhibit strong and diffuse CD34 immunoreactivity. (×100)

(B) Ki-67 low index (5% spindle cells show Ki67 immunoreactivity). (×100).

(C) Ki-67 high index (30% spindle cells show Ki67 immunoreactivity). (×100).

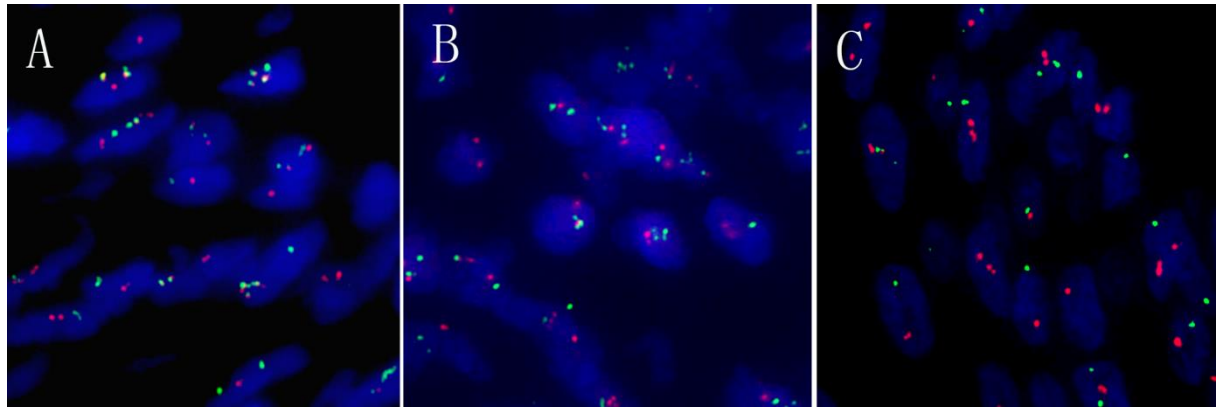


Figure 3 Detection of *COL1A1::PDGFB* fusion gene by fluorescence in site hybridization.

The red and green signals correspond to *PDGFB* and *COL1A1* respectively.

(A) Positive: Typical 1F/1R/1G fusion signal (yellow arrow) + remaining red-green separation signal

(B) Negative: Only red and green are separated, with no yellow fusion signal

(C) Negative control: Normal skin fibroblasts



(A) IGV View of *TPM3::NTRK1* Fusion

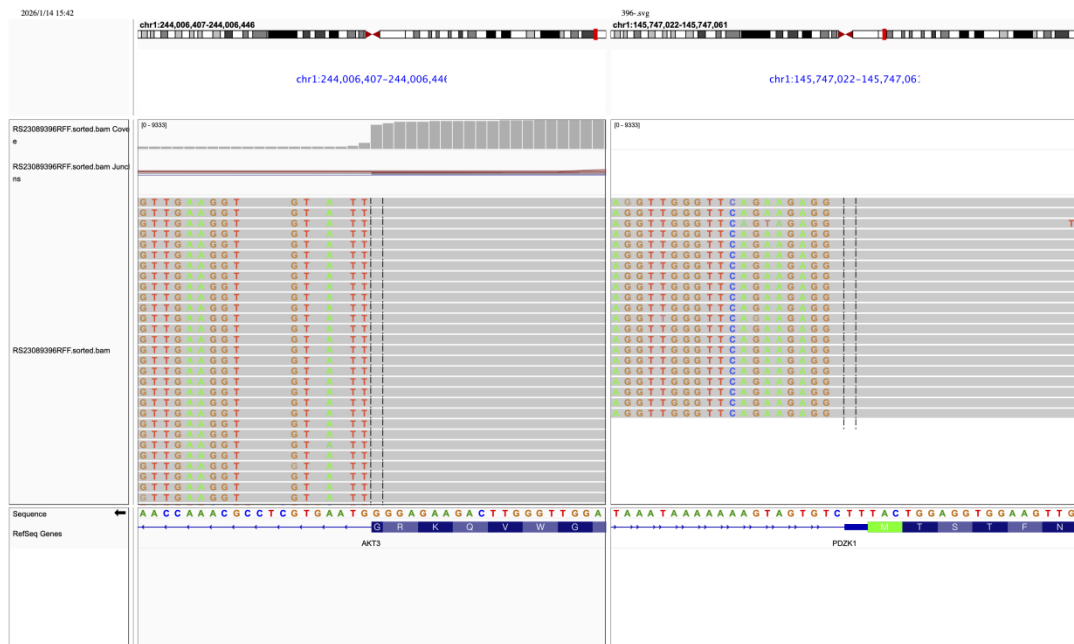
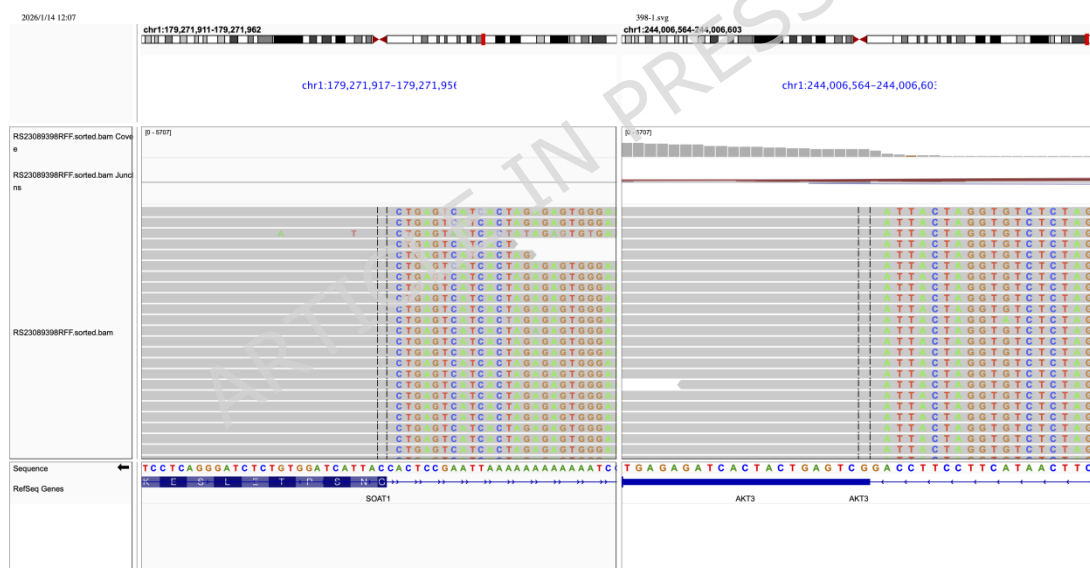
(B) IGV view of *AKT3::PDZK1* fusion(C) IGV view of *SOAT1::AKT3* fusion

Figure 4 Integrative Genomics Viewer (IGV) screenshots of paired-end RNA-seq reads showing the novel gene fusions. All coordinates are based on the hg19 reference genome.

- (A) *TPM3::NTRK1* fusion (chr1:154,142,876::chr1:156,844,363); breakpoint-spanning reads indicate fusion of TPM3 (NM_001043353.2) exon 7 to NTRK1 (NM_001007792.1) exon 10.
- (B) *AKT3::PDZK1* fusion (chr1:244,006,427::chr1:145,747,042); reads show fusion of AKT3 (NM_005465.7) exon 2 to PDZK1 (NM_002614.4) exon 3.
- (C) *SOAT1::AKT3* fusion (chr1:179,271,937::chr1:244,006,584); reads demonstrate fusion of SOAT1 (NM_001252511.1) exon 2 to AKT3 (NM_005465.7) exon 2.